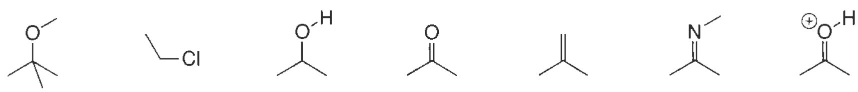
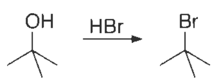
Practice Problems for Pushing Electrons

1. Identify any atoms, bonds, or lone pairs in the following molecules that could be considered as electron sources or sinks for two-electron arrows.

A. B. C. D. E. F. G.



2. Show mechanisms for the following transformations, along with all electron pushing. These reactions involve more than one step.

A.


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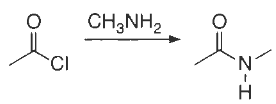
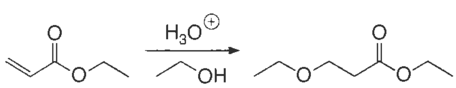
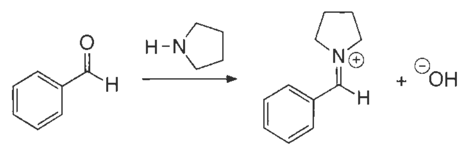
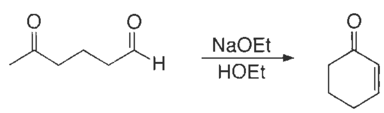
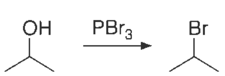


C.


D.


E.


3. The following are more complicated reactions than those presented to this point. However, they all involve two-electron steps that are similar to those presented in this book, especially in Chapters 10 and 11. Use your best chemical intuition to write reasonable mechanisms for these transformations. Draw all intermediates and show all electron pushing. The mechanisms that you write may not actually be the ones that have been supported by experiments. You cannot be expected to know this. However, your mechanism should be reasonable, and the electron pushing should be correct.

A.


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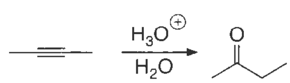
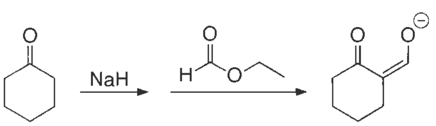


C.


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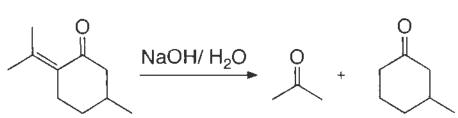
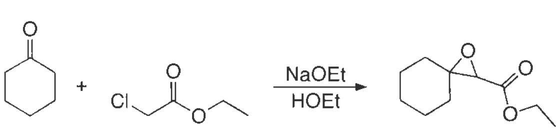
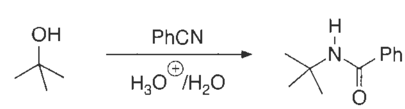
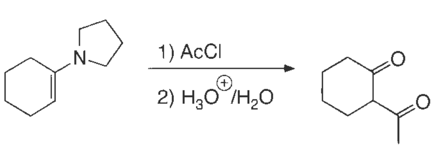


E.


F.


G.


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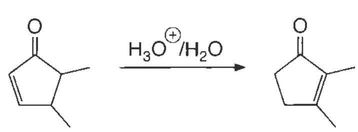
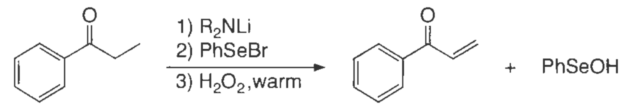
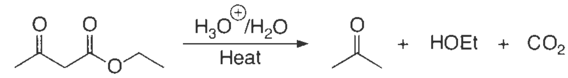


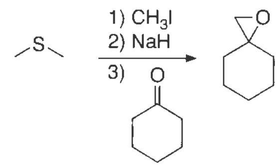
I.



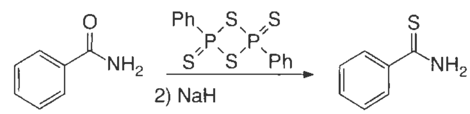
J.



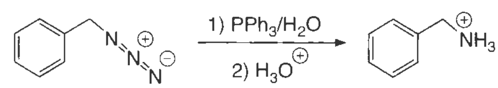
K.



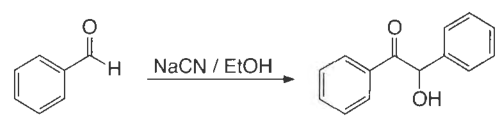
L.



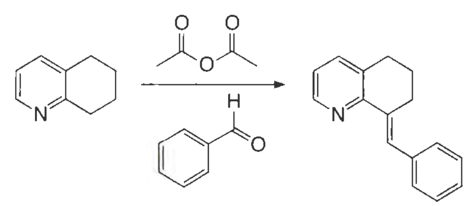
M.



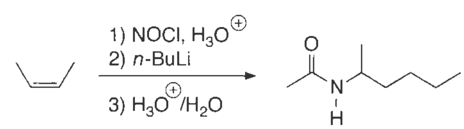
N.



O.

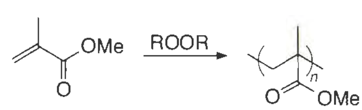


P.

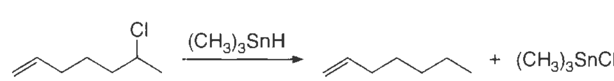


4. Write reasonable steps for the following radical reactions, showing all the proper electron pushing.

A.



B.



C.